

(Introduction to) Network Analysis 2025/26

Week	Date	Lectures	Labs	Coursework
1	Feb 16th	Course motivation, graph theory vs network science, graphology	<i>/ (lack of material)</i>	
2	Feb 23rd	Networkology, network representations & data, Erdos-Renyi model	Recitation on <i>NetworkX</i> library, <i>Pajek</i> format etc.	<i>Homework #1 out</i>
3	Mar 2nd	Configuration model, small-world networks & model, scale-free structure	Network representations, basic network algorithms	
4	Mar 9th	Scale-free networks & preferential attachment models, <i>course projects</i>	Advanced network algorithms, random graph models	
5	Mar 16th	Node position & measures of centrality, link analysis algorithms	Small-world & scale-free models, graphs vs networks	
6	Mar 23rd	Link importance & measures of bridging, networks in science, selected applications	Measures of node centrality, PageRank algorithm	<i>Homework #1 due</i>
7	Mar 30th	Community structure, community detection & graph partitioning	Node similarity, link betweenness, errors & attacks	<i>Homework #2 out</i>
8	Apr 6th	Node equivalence & blockmodeling, core-periphery structure	<i>/ (consultations)</i>	
	Apr	Node mixing in networks,	Community structure &	

9	13th	fragments & frequent subgraphs	detection, benchmark graphs	
10	Apr 20th	Network sampling & comparison, backbones & (convex) skeletons	Blockmodeling & block models, k -core network decomposition	
11	Apr 27th	<i>/ (vacation)</i>	<i>/ (vacation)</i>	<i>Homework #2 due</i>
12	May 4th	<i>/ (placeholder)</i>	Node mixing by (not) degree, graphlet degrees	
13	May 11th	Node layout & network visualization, selected applications, <i>centrality challenges</i>	Random-walk sampling, network comparison measures	
14	May 18th	Network inference & link prediction indices, graph machine learning	Board with pegs & bands, wiring diagrams, block models	
15	May 25th	Selected applications & research topics, course wrap-up, invited talks (tentative)	Node embeddings & classification, link prediction	<i>Course project due</i>
16	Jun 1st	<i>/ (semester ends)</i>	Selected analyses, review & examples of final exam	